

Unit 4 | Chapter 7: The Style (Advanced Layout & Responsive Design)

Learning Targets:	
❖ I can implement the CSS Box Model and correctly calculate rendered dimensions using content-box vs. border-box. ❖ I can build responsive, 1-dimensional (Flexbox) and 2-dimensional (Grid) layouts using fractional units (fr) and gap properties. ❖ I can execute a "Mobile-First" design strategy using media queries, viewport units (vw, vh), and the <meta name='viewport'> tag. ❖ I can precisely place elements using CSS Positioning (absolute, fixed, sticky) and manage stacking order with z-index. ❖ I can utilize modern CSS functions like calc(), clamp(), and minmax() to create fluid, accessible typography and layouts.	
Compelling Question:	
How does a "Layout Architect" use mathematical logic and responsive frameworks to ensure a website remains functional, organized, and beautiful for users like Olivia on any device?	
Score	Performance Indicators
4.0	I can: ☐ Design and build a 100% fluid layout using auto-fit vs. auto-fill and clamp() to ensure typography and grids scale perfectly without break-points. (WEB.4.3.5) ☐ Architect complex 2D layouts using grid-template-areas to visually map components to named strings. (WEB.5.2.1) ☐ Combine z-index and nested absolute positioning (relative to non-static ancestors) to create advanced UI components like modals or overlapping card systems. (WEB.5.2.5) ☐ Debug and resolve "spilling" content using overflow management and aspect-ratio properties to maintain media proportions during resize. (WEB.5.2.7)
3.5	In addition to score 3.0 performance, partial success at score 4.0 content.
3.0	I can: ☐ Correctly calculate total rendered width (Width + Padding + Border) and implement the "magic fix" of box-sizing: border-box. (WEB.5.2.6) ☐ Construct a Mobile-First stylesheet using min-width media queries to layer 

	styles for larger screens. (WEB.4.3.1) ☐ Implement Flexbox containers that utilize justify-content, align-items, and flex-wrap to space items evenly or across lines. (WEB.5.2.1) ☐ Use CSS Variables (:root and var()) and calc() to perform live math for layout dimensions. (WEB.5.2.2) ☐ Center block-level elements using margin: 0 auto and ensure images never overflow via max-width: 100%. (WEB.5.2.6)
2.5	No major errors or omissions regarding score 2.0 content, and partial success at score 3.0 content.
2.0	I can ☐ Define common terminology and identify syntax: (WEB.4.2.1) ☐ Box Model, Margin Collapse, rem vs. em, vw/vh, fr unit, sticky/fixed/absolute, z-index, justify-content vs. align-items, flex-direction, grid-column: span, media queries, display: none vs. visibility: hidden, and :hover.
1.5	Partial success at score 2.0 content, and major errors or omissions at 3.0.
1.0	With help, partial success at score 2.0 and score 3.0 content.
0.5	With help, partial success at score 2.0 but not at score 3.0 content.
0.0	Even with help, no success or attempt made.